



From Inbox to Board

AI & Automation Officer · Part 1
AI-Powered Email-to-ClickUp Automation

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Focus: solution design · agent workflow · MCP integration · AWS architecture

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1. Problem & Objective

The current process starts with work requests arriving by email. A person reads each message, decides whether it is actually a request that should be tracked, and then manually creates or updates a ClickUp task. The process creates avoidable manual effort and introduces the risk of missed requests, duplicate work, and incorrect urgency or routing.

1.1 Objective

Reduce the manual effort required to turn incoming work-request emails into actionable ClickUp tasks, while keeping human judgment in the loop for ambiguous, incomplete, duplicate, or sensitive requests.

2. Proposed Solution

The proposed solution is an Inbox Reviewer Agent built with the Strands Agents framework and hosted on Amazon Bedrock AgentCore Runtime. The agent reads incoming email context, determines the appropriate next action, extracts the information required for execution, and uses authorized MCP tools to interact with ClickUp or Outlook.

The design intentionally treats automation as an action-selection problem rather than a simple “is this a task?” classifier. The agent chooses one of five outcomes: CREATE_TASK, UPDATE_TASK, REPLY, IGNORE, or HUMAN_REVIEW.

Task field extraction: For CREATE_TASK and UPDATE_TASK actions, the agent extracts or derives the ClickUp task fields: title, description, assignee, priority, due date, and status. Fields that cannot be determined reliably are left for human review rather than guessed.

2.1 Action Model

Action	When it is used	Expected result
CREATE_TASK	Clear, actionable work request with enough information.	Create a new ClickUp task.
UPDATE_TASK	Request refers to existing tracked work or needs an existing task changed.	Find and update the relevant ClickUp task.
REPLY	The correct response is communication rather than task creation.	Draft a reply through Outlook; send it only when the configured policy allows automatic sending, otherwise route it for human approval.
IGNORE	Informational or conversational email requiring no action.	No task and no further action.
HUMAN_REVIEW	Ambiguity, missing critical data, possible duplicate, or sensitive action.	Present proposed action to a reviewer for approval, edit, or rejection.

3. High-Level Business Workflow

At the business level, the solution follows a simple pattern: receive the email, understand intent, decide what should happen next, execute routine actions automatically, and route exceptions to a person.

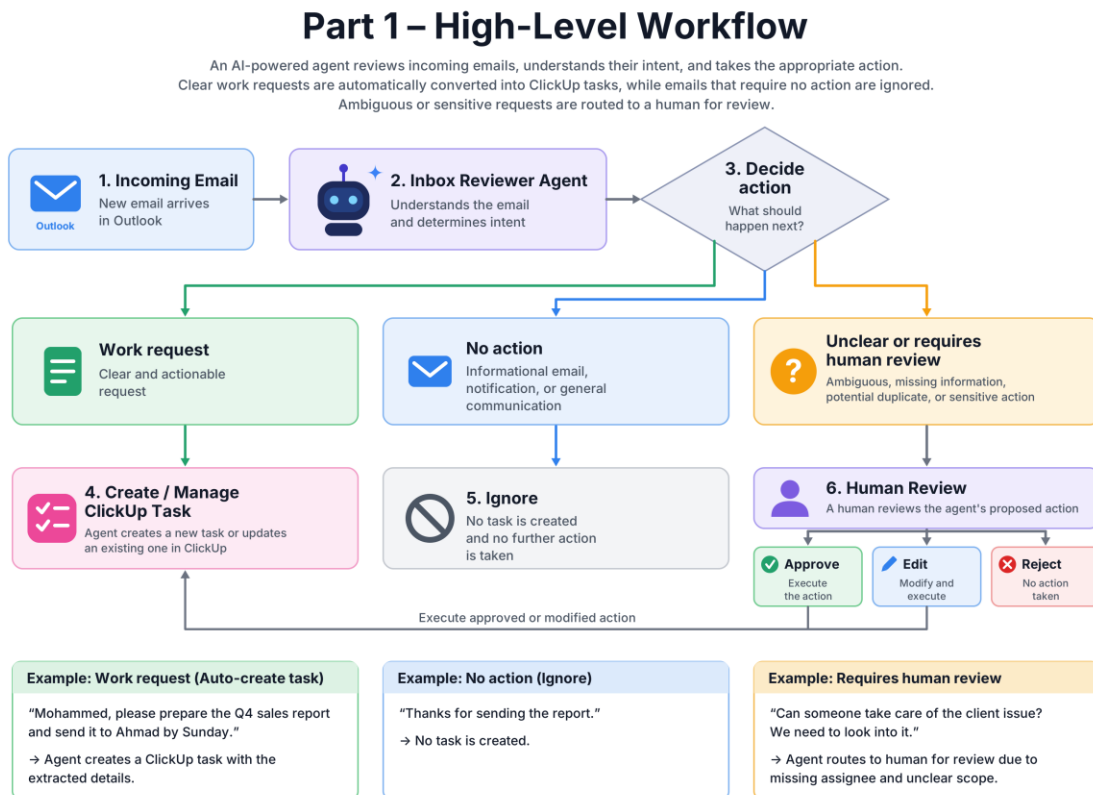


Figure 1: High-Level Business Workflow

4. Technical Workflow: Agent + MCP

The technical workflow separates reasoning from integration. The Strands agent decides what action is appropriate, while AgentCore Gateway exposes the external capabilities through MCP. This keeps business reasoning independent from the specific APIs used by ClickUp and Outlook.

Part 1 – Technical Workflow (Agent + MCP)

Detailed workflow showing how the Inbox Reviewer Agent processes emails, uses MCP tools, and involves human review when needed.

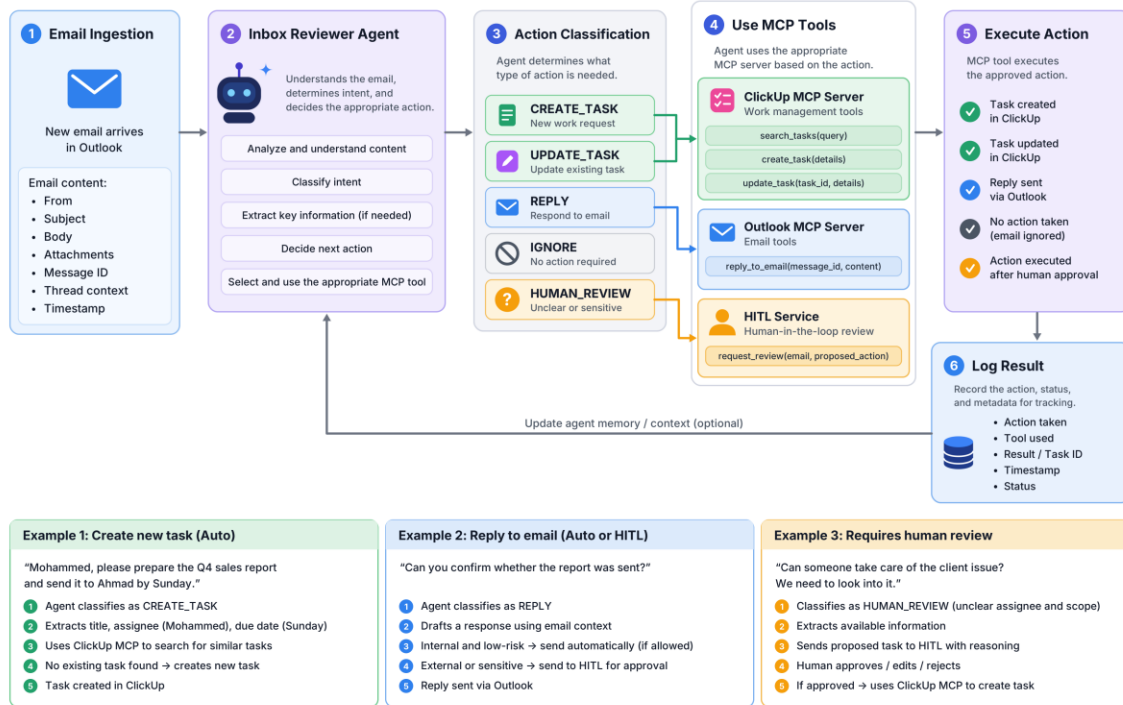


Figure 2: Technical Workflow (Agent + MCP). Detailed execution flow showing email context, agent decisions, MCP tool selection, human review, execution, and result logging.

4.1 Tool Boundaries

- **ClickUp MCP:** search_tasks(query), create_task(details), update_task(task_id, details).
- **Outlook MCP:** reply_to_email(message_id, content).
- **Human-in-the-loop:** request or trigger review when the proposed action should not be executed automatically.

The agent should receive only the minimum tool access required for this workflow, even where a broader MCP connector exposes additional capabilities.

5. AWS Architecture

The AWS design uses managed services to keep the solution serverless and focused while still providing clear boundaries for agent execution, tool integration, human review, state, security, and observability.

Architecture rationale: The solution is designed as a production-oriented cloud workflow rather than a local or purely low-code automation. Managed AWS services provide clear security boundaries, scalable execution, centralized observability, controlled access to external tools, and lightweight persistent state. This allows the proof of concept to evolve toward a production deployment without redesigning the core workflow.

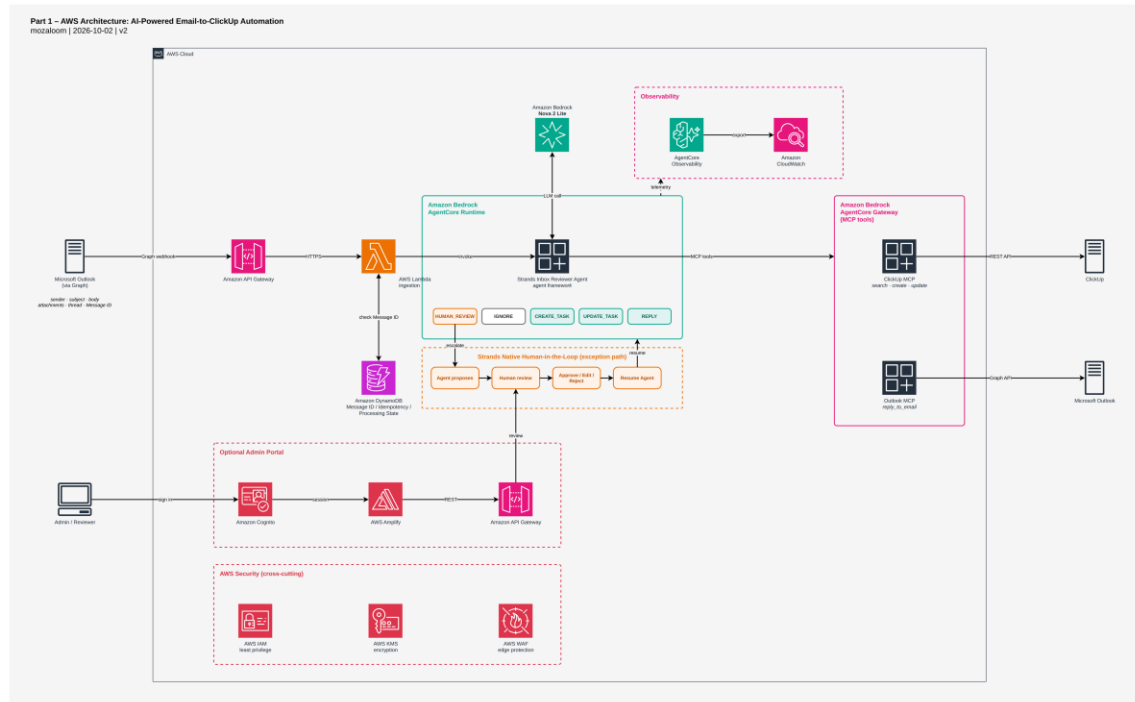


Figure 3: AWS Architecture. Proposed AWS architecture for the AI-powered email-to-ClickUp automation.

5.1 Core Components

Component	Role
Amazon Bedrock AgentCore Runtime	Hosts the Strands Inbox Reviewer Agent and provides the managed runtime boundary for agent execution.
Strands Agents	Provides the agent framework and control flow used to interpret email context and select an action.
Amazon Bedrock: Nova 2 Lite	Default model choice for routine email classification, extraction, and business-process reasoning.
Amazon Bedrock AgentCore Gateway	Provides the MCP tool layer used by the agent to reach ClickUp and Outlook integrations.
API Gateway + Lambda	Receives the Microsoft Graph/Outlook webhook path and performs lightweight ingestion before invoking the agent.
Amazon DynamoDB	Stores message IDs and lightweight processing state to support idempotency and avoid duplicate processing.
Strands Native HITL	Pauses execution when human judgment is required and resumes after approval, edit, or rejection.
AgentCore Observability + CloudWatch	Provides execution visibility across agent, model, tool, and error paths.

5.2 Optional Components

- **Amazon Cognito + AWS Amplify + API Gateway:** optional admin/reviewer portal for managing pending human-review items.
- **AWS WAF:** protects exposed API or web endpoints when a public surface is introduced.
- **AWS KMS:** encryption/key management for stored sensitive information where required.

6. Automation & Human Review Strategy

The system is designed so that automation is the default for clear, low-risk requests, while human review acts as an exception mechanism. The agent should not create work simply because an email contains an instruction-like phrase; it should assess whether the request is actionable and sufficiently specified.

Automation boundary: The system automatically executes only actions where the intended action, required fields, target, and risk are sufficiently clear. If any of these are uncertain, the system does not guess; it requests human review. In short: clear, low-risk requests are automated; unclear or sensitive ones go to a person.

6.1 Automatic Processing

- Clear task request with identifiable title, scope, and enough task metadata → create or update ClickUp.
- Clear, low-risk response request with known context and acceptable risk → prepare a reply and send it only if the configured policy allows automatic sending.
- Informational or conversational email with no action required → ignore.

6.2 Human Review Triggers

- Ambiguous request or unclear intended action.
- Missing critical information such as assignee or required scope.
- Potential duplicate where the correct existing task cannot be identified confidently.
- External communication, or any other sensitive or externally visible action, unless the configured policy explicitly allows automatic sending.

HITL interaction

The reviewer sees the email context and the agent's proposed action, then chooses Approve, Edit, or Reject. An approved or edited action resumes the agent workflow and uses the same authorized MCP layer for execution.

7. Duplicate Prevention & State

Duplicate prevention operates at two levels: message-level idempotency prevents repeated delivery of the same email, while ClickUp task search helps identify existing work that may represent the same request even when it originated from a different email.

Message-level idempotency: The ingestion layer records the message ID and processing status before or during workflow execution. A repeated delivery of the same message can therefore be detected and prevented from creating duplicate ClickUp work.

Task-level search: Before creating a task, the agent searches ClickUp for similar existing work. If a matching task is identified, the request is routed to UPDATE_TASK instead of creating a new one; if a possible match cannot be identified confidently, it is routed to HUMAN_REVIEW.

DynamoDB is used only for lightweight application state; ClickUp remains the system of record for business tasks.

8. Example Scenarios

Example 1: Clear work request

"Mohammed, please prepare the Q4 sales report and send it to Ahmad by Sunday."

CREATE_TASK → extract title, assignee, due date → search for similar task → create only if no existing task is found.

Example 2: Informational email

"Thanks for sending the report."

IGNORE → no task created and no further action taken.

Example 3: Ambiguous request

"Can someone take care of the client issue? We need to look into it."

HUMAN_REVIEW → proposed action is shown to a reviewer because assignee and scope are unclear.

Example 4: Existing work

An email asks to change the due date or scope of a task that already exists.

UPDATE_TASK → search ClickUp → update the matching task.

Example 5: Duplicate delivery

The same email is delivered more than once.

Idempotency check → detect the existing message ID → prevent duplicate processing.

Example 6: Sensitive external action

"Please send the client an email confirming the revised delivery date."

HUMAN_REVIEW → external communication requires approval before sending.

9. Alternative Approaches Considered & Rejected

Alternative 1: Deterministic Rule-Based Email Automation

A simpler approach would use keyword matching, predefined templates, and fixed conditions to determine whether an email should become a ClickUp task.

We decided against using this as the primary approach because email requests vary significantly in wording, context, completeness, and intent. As the number of request types grows, maintaining

rules becomes increasingly difficult. A rule-based solution is also weaker at handling ambiguity, extracting structured task fields, and identifying potential duplicates.

The proposed agentic approach provides more flexibility in understanding natural-language requests while keeping execution constrained through explicit actions, MCP tools, idempotency controls, and human review.

Alternative 2: Basic Low-Code Automation Platform

We also considered implementing the workflow using a basic low-code automation platform with predefined triggers, conditions, and ClickUp actions.

This could reduce initial development effort, but it provides less flexibility for contextual reasoning, ambiguity handling, duplicate detection, and controlled human-in-the-loop decisions. It can also become harder to govern as the workflow evolves and more systems and actions are introduced.

Instead, we designed the solution around a production-oriented cloud architecture. This better reflects how the solution could evolve beyond a proof of concept, with centralized identity and access control, auditability, observability, security controls, scalability, and the ability to accommodate enterprise data and compliance requirements.

Why this approach was selected

The goal was not simply to automate task creation, but to create a workflow that can reliably understand incoming requests, choose the appropriate action, and escalate uncertain or sensitive cases to a human while being designed with production deployment considerations in mind.

10. Security, Reliability & Observability

10.1 Security

- Least-privilege IAM roles for the agent runtime, integration tools, and supporting services.
- Cognito for authenticated reviewer/admin access when the optional portal is used.
- KMS for encryption/key management where persistent sensitive data requires customer-managed keys.
- WAF for exposed API/web entry points when a public endpoint is introduced.

10.2 Reliability

- Idempotency based on message ID to reduce duplicate task creation.
- Structured action outputs to make agent decisions machine-checkable and easier to validate.
- Human review for cases where confidence or completeness is insufficient for safe automation.
- MCP tool boundaries limit the agent to the actions it is authorized to perform.
- **Failure handling:** If an MCP tool call, ClickUp operation, or Outlook operation fails, the system records the failure and does not silently retry an action that could create duplicate work. Retryable operations can be retried using idempotency controls; persistent failures are surfaced for human review or operational intervention.

10.3 Observability

- AgentCore Observability and CloudWatch provide visibility into agent execution, model calls, tool calls, errors, latency, and outcomes.
- Store result metadata such as action, status, task ID, timestamps, and processing state for operational traceability.

11. Part 1 Summary

The proposed solution turns an email-heavy manual process into a controlled AI-assisted workflow. The Inbox Reviewer Agent handles interpretation and action selection, MCP provides a consistent integration boundary, ClickUp remains the destination for tracked work, and human review protects the process when information is incomplete or risk is higher.

Design principle

Automate the routine, constrain the tools, make uncertainty visible, and keep a human in control of exceptions.